



IT582: Foundations of Machine Learning

Lab-1: Python Foundations and Linear Regression

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Submission Instructions

Upload **only one Jupyter Notebook (.ipynb)** file to Moodle.
The notebook should contain all code, outputs, plots and conclusions.
Name the notebook as:

[studentid]_lab1.ipynb

Any other filename or additional files will not be accepted.

Part 1: Python Foundations for Machine Learning

Task 1: Python and NumPy

Complete the following using Python and NumPy:

1. Create a list containing numerical values and remove all negative values using list comprehension.
2. Write a function to calculate the mean, minimum and maximum of a list.
3. Create a NumPy array of size 5×4 and perform:
 - Array slicing
 - Boolean indexing
 - Row-wise and column-wise mean
 - Matrix multiplication
4. Calculate Mean Squared Error using:
 - A Python loop
 - A vectorized NumPy operation
5. Compare the execution time of both methods.

Task 2: pandas and Matplotlib

Load the **California Housing Dataset** using:

```
fetch_california_housing(as_frame=True)
```

Perform the following:

1. Explore the dataset by performing the following:
 - Display the first five rows.
 - Display the number of rows and columns.
 - Display the names of all columns.
 - Display the data type of each column.
 - Display the summary statistics of all numerical columns.
2. Check whether the dataset contains:
 - Missing values
 - Duplicate rows
3. Create the following plots:
 - A histogram showing the distribution of `MedHouseVal`.
 - A scatter plot with `MedInc` on the x-axis and `MedHouseVal` on the y-axis.
 - A heatmap showing the pairwise correlation between all features.

Write observations for each plot.

Part 2: Linear Regression from Scratch

Dataset

Use the **California Housing Dataset** available in `sklearn.datasets`. The dataset should be loaded using:

```
fetch_california_housing()
```

Scikit-learn may only be used for loading the dataset. The Linear Regression model and loss function must be **implemented from scratch**.

Task 3: Dataset Preparation

1. Split the dataset into:
 - Training Set (60%)
 - Validation Set (20%)
 - Test Set (20%)
2. Standardize the input features using only the training-set mean and standard deviation.
3. Add a column of ones to represent the bias parameter.

Task 4: Implement Linear Regression

Implement Linear Regression from scratch using NumPy and Batch Gradient Descent. In Batch Gradient Descent, the entire training set is used to compute the gradient before updating the model parameters once per epoch.

The Mean Squared Error should be implemented as:

$$J(\theta) = \frac{1}{m} \sum_{i=1}^m (\hat{y}_i - y_i)^2$$

Task 5: Learning Rate Analysis

Train the implemented model with each of the following learning rates:

$$10^{-5}, 10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}, 1$$

Plot:

- Epoch vs Training MSE
- Epoch vs Validation MSE

Analyse the plots and write conclusions

Task 6: Parameter vs MSE Loss

For this task, use only the `MedInc` feature and the target variable. Train a univariate Linear Regression model:

$$\hat{y} = \theta_0 + \theta_1 x$$

Perform the following:

1. Keep θ_1 fixed and vary θ_0 over a suitable range.
2. Plot θ_0 vs MSE loss.
3. Keep θ_0 fixed and vary θ_1 over a suitable range.
4. Plot θ_1 vs MSE loss.
5. Mark the learned parameter values on both plots.

Write a conclusion explaining:

- What the minimum point of the curve represents.
- Why the loss increases when the parameter moves away from the optimum.
- How Gradient Descent moves towards the minimum.

Task 7: Inference on Test Set

Using the best model obtained from the previous tasks:

1. Evaluate the model on the unseen test set.
2. Report training, validation and test MSE.
3. Plot Actual vs Predicted values for the test set.

Write observations.